Task-10:- Requirement and Analysis

1. Introduction

1.1 Purpose

The purpose of this document is to describe the requirements and analysis for creating a **custom AWS CloudWatch dashboard** that monitors multiple EC2 instances within the AWS account of ABC Technologies.

This document is intended for:

- DevOps engineers
- Cloud administrators
- Operations team members
- Project supervisors and stakeholders

It provides a clear understanding of the system requirements, functional behavior, and monitoring objectives.

1.2 Scope

The scope of this project includes the design and creation of a **centralized AWS CloudWatch Dashboard** to monitor the performance and health of **four EC2 instances**.

The dashboard will display infrastructure metrics such as:

- CPU Utilization
- Memory Usage (via CloudWatch Agent)
- Disk Usage
- Network In
- Network Out

The dashboard will present these metrics using different visualization types such as:

- Line Charts
- Pie Charts
- Histograms

Included in Scope

- Monitoring of four EC2 instances

- Infrastructure metrics visualization
- CloudWatch dashboard configuration
- Dashboard layout design
- Performance comparison between instances

Not Included in Scope

- Application-level monitoring
- Automated alerts and notifications
- Cost monitoring
- Infrastructure automation scripts

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
AWS	Amazon Web Services
EC2	Elastic Compute Cloud
CloudWatch	AWS monitoring and observability service
CPU	Central Processing Unit
RAM	Random Access Memory
Dashboard	Visual interface for monitoring metrics
DevOps	Development and Operations practices
Metrics	Numerical measurements representing system performance

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1.4 References

The following resources were used while preparing this document:

- AWS Official Documentation – CloudWatch
  https://docs.aws.amazon.com/cloudwatch/

- AWS EC2 Monitoring Documentation
  Monitor Amazon EC2 resources - Amazon Elastic Compute Cloud
- AWS CloudWatch Dashboard Guide
  Using Amazon CloudWatch dashboards - Amazon CloudWatch

1.5 Overview

This document is structured as follows:

- Section 1 introduces the purpose and scope of the project.
- Section 2 describes the overall project objectives and business context.
- Section 3 outlines the functional requirements and use cases.
- Section 4 explains the non-functional requirements.
- Section 5 discusses assumptions and dependencies.
- Section 6 presents risk analysis and mitigation strategies.
- Section 7 provides a glossary of important terms.
- Section 8 contains additional supporting information.
- Section 9 includes customer acknowledgement.

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2. Project Overview

2.1 Project Objectives

The primary objective of this project is to develop a **centralized monitoring dashboard using AWS CloudWatch** to monitor multiple EC2 instances.

The dashboard should:

- Provide a clear view of system performance
- Monitor infrastructure health
- Allow comparison between EC2 instances
- Help identify abnormal resource usage

2.2 Business Goals

The business goals of this project include:

- Improving infrastructure monitoring efficiency
- Providing a centralized monitoring interface

- Reducing the time required to check individual EC2 instances
- Improving operational visibility
- Supporting proactive infrastructure management

2.3 Background

ABC Technologies currently operates multiple EC2 instances for application workloads and testing environments.

The operations team currently checks instance metrics individually using the AWS console. This process is inefficient and makes it difficult to quickly identify performance issues across multiple systems.

To solve this problem, the organization requires a **centralized CloudWatch dashboard** that displays important infrastructure metrics in a visual format.

This solution will simplify monitoring and allow the operations team to quickly understand system health.

2.4 Constraints

The following constraints may affect the project:

Time Constraint

The task must be completed within **3 hours**.

Technical Constraints

- Default EC2 monitoring does not provide memory usage metrics.
- Memory and disk metrics require installation of the **CloudWatch Agent**.

AWS Permissions

The user must have appropriate permissions to create dashboards and access CloudWatch metrics.

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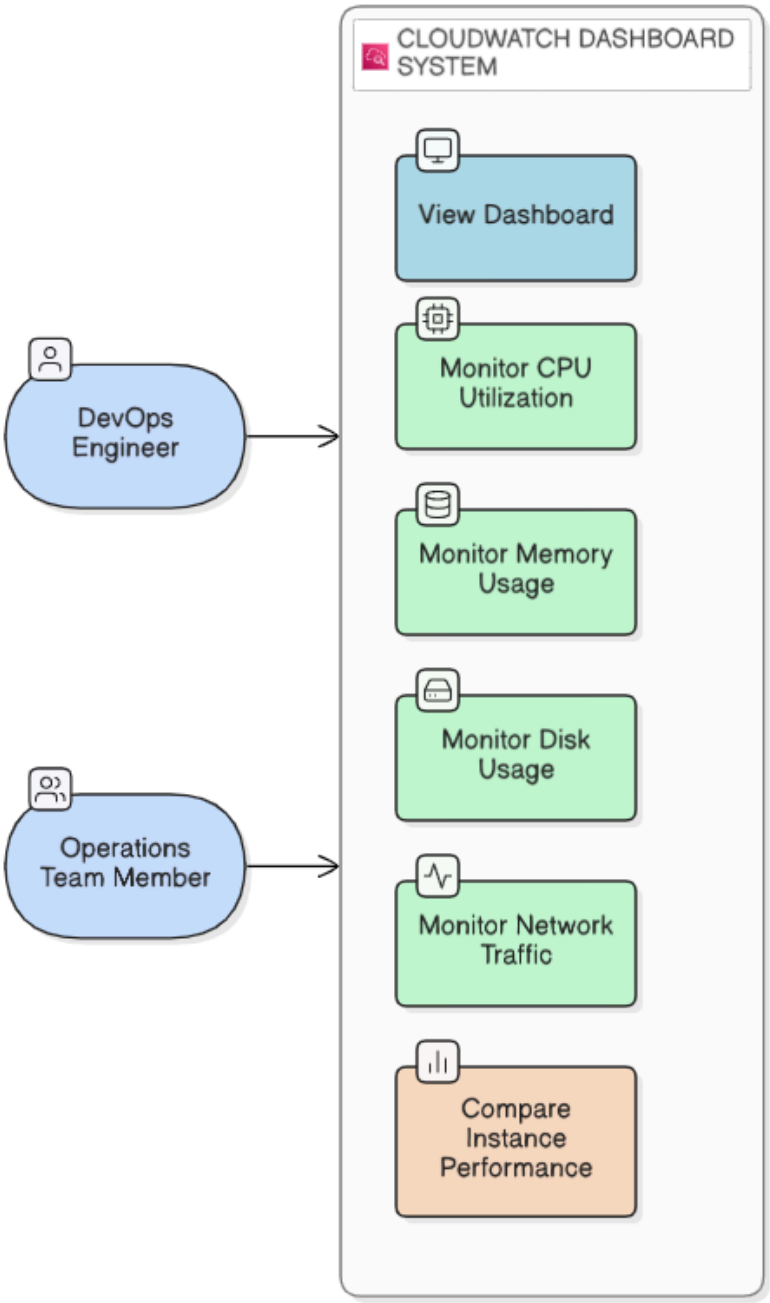
3. Functional Requirements

3.1 Use Case Diagram

The main actor in the system is the **DevOps Engineer / Operations Team Member**.

Use cases include:

- View CloudWatch Dashboard
- Monitor CPU Utilization
- Monitor Memory Usage
- Monitor Disk Usage
- Monitor Network Traffic
- Compare EC2 Instance Performance



3.2 Use Cases

3.2.1 Use Case 1: View Infrastructure Dashboard

Description:
The user views a centralized CloudWatch dashboard that displays infrastructure metrics for all EC2 instances.

Actors:
DevOps Engineer
Operations Team Member

Preconditions:

- AWS account access is available
- CloudWatch dashboard has been created
- EC2 instances are running

Postconditions:

- The user can view the health and performance of EC2 instances.

Flow of Events:

1. User logs into AWS Management Console.
2. User navigates to CloudWatch service.
3. User opens the custom CloudWatch dashboard.
4. Dashboard displays metrics for all EC2 instances.

Alternative Flows:

- Dashboard fails to load due to insufficient permissions.
- Metrics are unavailable due to monitoring configuration issues.

3.2.2 Use Case 2: Monitor CPU Utilization

Description:
The user monitors CPU utilization across all EC2 instances.

Actors:
DevOps Engineer

Preconditions:

- EC2 monitoring is enabled.

Postconditions:

- CPU usage trends can be analyzed.

Flow of Events:

1. User opens the dashboard.
2. CPU utilization widget displays a line chart.
3. The user observes CPU usage trends for each instance.

Alternative Flows:

- CPU metrics are delayed due to monitoring frequency.

3.2.3 Use Case 3: Compare Instance Performance

Description:
The user compares performance metrics across multiple EC2 instances.

Actors:
Operations Team

Preconditions:

- All instances are included in the dashboard.

Postconditions:

- Performance comparison is visible.

Flow of Events:

1. User views dashboard widgets.
2. Metrics for multiple instances are displayed.
3. User compares usage patterns between instances.

Alternative Flows:

- One instance may show abnormal usage.

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4. Non-Functional Requirements

4.1 Performance Requirements

- Dashboard should load within **5 seconds**.
- Metrics should update regularly based on CloudWatch refresh intervals.

4.2 Security Requirements

- Only authorized AWS users should access the dashboard.
- AWS IAM roles and permissions should control dashboard access.

4.3 Usability Requirements

- Dashboard layout should be simple and readable.
- Widgets should be clearly labeled.
- Visualizations should help quickly understand system performance.

4.4 Reliability Requirements

- CloudWatch should consistently collect metrics.
- Dashboard should remain available whenever the AWS console is accessible.

4.5 Compliance Requirements

The solution should follow **AWS security and monitoring best practices**.

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5. Assumptions and Dependencies

5.1 Assumptions

- All EC2 instances are running in the same AWS account.
- CloudWatch monitoring is enabled.
- The user has sufficient AWS permissions.

5.2 Dependencies

The project depends on:

- AWS CloudWatch service

- AWS EC2 service
- CloudWatch Agent (for memory and disk metrics)
- Internet access for AWS console

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6. Risk Analysis

6.1 Risk Identification

Risk	Description
Missing Metrics	Some metrics like memory may not be available
Incorrect Dashboard Layout	Poor visualization may reduce readability
Permission Issues	User may lack permissions to access metrics

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6.2 Risk Mitigation

Risk	Mitigation
Missing Metrics	Install CloudWatch Agent
Layout Issues	Use structured widget layout
Permission Issues	Configure IAM policies properly

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7. Glossary

Term	Meaning
CloudWatch	AWS monitoring service
EC2	Virtual machine service in AWS
Dashboard	Visual monitoring interface
Metrics	Data used to measure performance

8. Appendices

Additional materials may include:

- CloudWatch dashboard screenshots
- Dashboard configuration details
- Monitoring architecture diagrams

9. Customer Acknowledgement

Name: _____

Title: _____

Signature: _____

Date: _____